

Resume

Kanu Madhok

952-303-1045 · madhok.kanu@gmail.com · github.com/kmadhok ·
linkedin.com/in/kanu-madhok

End-to-end AI engineer who embeds with stakeholders, rapidly prototypes, and ships production-grade agentic systems with measurable business impact.
Live Demo: NL-to-SQL Copilot — Dockerized Streamlit on GCP Cloud Run ·
<https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

EXPERIENCE

Walmart Data Ventures — Senior Data Analyst

Sep 2024 – Present · Chicago, IL

- Built and shipped to production an autonomous Jira ticket-resolution agent that closed 400+ data requests across 8 categories over 15 months, cutting routine turnaround from 30–60 minutes to under 10 minutes per request. Removed ~130–330+ hours of request-cycle time on routine tickets alone. Under the hood: 6-gate triage, MCP-driven context gathering, agentic RAG with FAISS + BM25 retrieval over 11,000 historical SQL queries, and self-healing execution with a 10-retry budget. Stakeholders never knew an agent was answering them.
- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables with sub-agent orchestration and human-in-the-loop validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate); 8 deterministic SQL rules and a 23-case golden-query evaluation suite. Partnered with Product, Data Science, and business stakeholders to translate ambiguous business questions into agent-ready specs. Approved by Sr. Director and product leadership and currently the only AI skill in active use by business teams on Data Ventures.
- Built a hybrid orchestrator on top of the analytics agent that lets Product self-serve analytics requests they previously submitted as tickets. Routes questions through two backends in parallel — a CubeJS semantic-layer skill co-built with the DS and product team, and the context-engineered agent itself — compares outputs, reasons about disagreements, and feeds discrepancies back to refine semantic-layer definitions over time. Converted 30–60 minutes of analyst execution per covered request into product-led self-service.
- Built and deployed to production a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; end-to-end observability via decision logging and BigQuery state persistence. Designed a 3-tier event-driven v2 architecture self-configuring through a BigQuery event registry so new event types ship without code deploys.
- Replaced ~20 recurring recruitment workflows run by hand 1–2x weekly with end-to-end automation on a 68-table BigQuery platform; stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates, reducing a recurring 3-hour task to ~3 minutes of setup.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes. Making a Difference Award.

FTI Consulting — UChicago MSADS Capstone

Jan – Aug 2024 · Chicago, IL

- Built a RAG-to-proposal generator deployed on AWS EC2 and adopted into FTI Consulting's workflow; cut proposal time ~30%. Best in Show — UChicago MSADS Capstone.

University of Chicago, Data Science Institute

Jan – Sep 2024 · Chicago, IL

- Built a persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

Innovare (EdTech Startup) — Data Scientist

Oct 2022 – Mar 2023 · Chicago, IL

- Built a logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures -15%.

SELECTED PROJECT

- **Live NL-to-SQL Copilot** — Streamlit + Gemini 2.5 + LangChain-orchestrated RAG (embeddings over historical queries), BigQuery execution, auto-visualization. Public-facing version of the agent-on-data architecture deployed inside Walmart Data Ventures. Demo: <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Agent stack: MCP server design, sub-agent orchestration, context engineering, prompt engineering, RAG (embeddings + FAISS + BM25 + RRF), LangChain, ChromaDB semantic memory, evaluation suites (golden sets, human-in-the-loop validation), multi-model routing across providers, headless cron agents, event-driven architectures, decision logging / observability

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

AI-assisted development: Claude Code, Cursor

APIs / triggers / glue: FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores, CI/CD (pytest, version control)

Data: BigQuery, dbt, SQL, Python, PySpark, Airflow, Databricks, MySQL

Viz / front-end: Power BI (DAX, XMLA TMSL), Tableau, LookML, Streamlit, Matplotlib, React Recharts